Zaheed Bhaiji (he/him)

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EDUCATION

University of British Columbia May 2029

Bachelor of Applied Science – Electrical Engineering

- Coop Availability: 4, 8, 12, or 16 months
- Relevant Courses: CPSC 259 (Data Structures and Algorithms for Electrical Engineers), CPEN 211 (Computing Systems I), ELEC 201 (Circuit Analysis I)

TECHNICAL SKILLS

Lab & Hardware: Circuit Debugging, Oscilloscopes, Function Generators, DMMs, DC Power Supplies, Microcontrollers (ESP32, STM32), FPGA, Sensor Interfacing, Protocols (UART, I2C), Breadboarding, Soldering
Languages & Simulation: C/C++, SystemVerilog, Assembly (RISC-V), Python, FreeRTOS, STM32CubeIDE, Quartus Prime, Questa/ModelSim, MATLAB
CAD & Prototyping: KiCad, Altium Designer, Onshape, SolidWorks, FDM 3D Printing, Waterjet Machining

ENGINEERING DESIGN TEAMS

Uncrewed Aircraft Systems | Vancouver, BC January 2026 – Present

Avionics & Electrical Systems – Payload Board

- Engineering a control board end-to-end, spanning schematic design, PCB layout, benchtop debugging and firmware to control a payload claw mechanism with sensors and actuators
- Organizing cross-disciplinary meetings with the payload team to configure hardware according to task specifications and budget constraints

Aircraft Design – Butterfly Drone

- Designed and manufactured a custom training drone in Onshape, implementing an open-plane modular chassis to configure different payloads or peripherals
- Integrated crossbars into frame geometry for torsional rigidity and component mounting space.
- Hand-soldered brushless motors to ESCs, routed signal lines to flight controller pads

TECHNICAL PROJECTS

Robot Arm (Embedded Systems PCB) | Personal Project August 2026 – Present

- Engineered and tested 2-layer PCB in KiCad for a bare-metal ESP32 SoC, with custom boot/flashing circuitry and IP2326 boost charger IC to manage 2S LiPo batteries
- Employed a multi-rail power system with a switching buck converter and LDOs to step voltage down to 5V power and 3.3V logic, calculating trace widths to handle current and prevent overheating
- Programmed FreeRTOS firmware in C with priority levels to read 4 IMUs over I2C, calculate relative joint angles and drive a 5-DOF arm via PWM signals

Custom 8-Bit Processor | Coursework Project August 2026 – Present

- Synthesized an 8-Bit CPU in System Verilog for a DE10-Lite FPGA, integrating a 4-function ALU with condition flags and memory mapped I/O to execute and display basic math algorithms (Fibonacci, factorials, GCD)

WORK EXPERIENCE

Telus Digital | Langley, BC July 2026 – Present

Online Data Analyst

- Evaluated search queries and machine learning outputs, checking for response relevance and training data accuracy across high-volume datasets

VOLUNTEER EXPERIENCE

Medical Envoys for Community Health | Langley, BC July 2023 – May 2025

Nonprofit Founder

- Directed a 30+ member student-based organization to spread awareness for local health quality
- Hosted major community outreach events, raising over $3000 to support community hospitals